The Time Value of Money

4.1 The Timeline

4.2 The Three Rules of Time Travel

4.3 Valuing a Stream of Cash Flows

4.4 Calculating the Net Present Value

4.5 Perpetuities, Annuities, and Other Special Cases

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Chapter Outline (cont'd)

4.6 Solving Problems with a Spreadsheet Program

4.7 Solving for Variables Other Than Present Value or Future Value

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4.1 The Timeline

- A timeline is a linear representation of the timing of potential cash flows.
- Drawing a timeline of the cash flows will help you visualize the financial problem.

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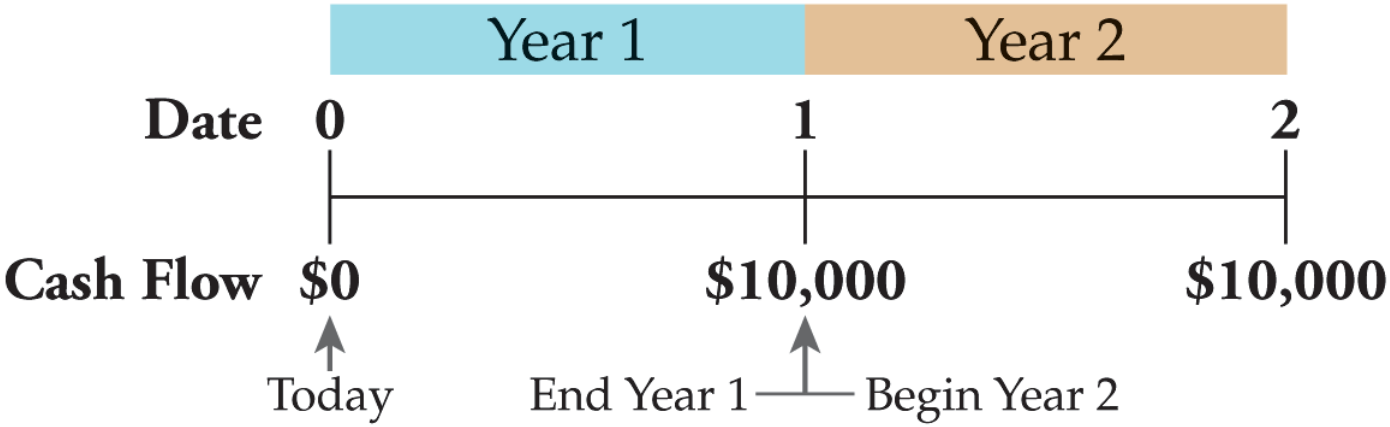
4.1 The Timeline (cont'd)

- Assume that you made a loan to a friend. You will be repaid in two payments, one at the end of each year over the next two years.



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4.1 The Timeline (cont'd)

- Differentiate between two types of cash flows
  - Inflows are positive cash flows.
  - Outflows are negative cash flows, which are indicated with a – (minus) sign.

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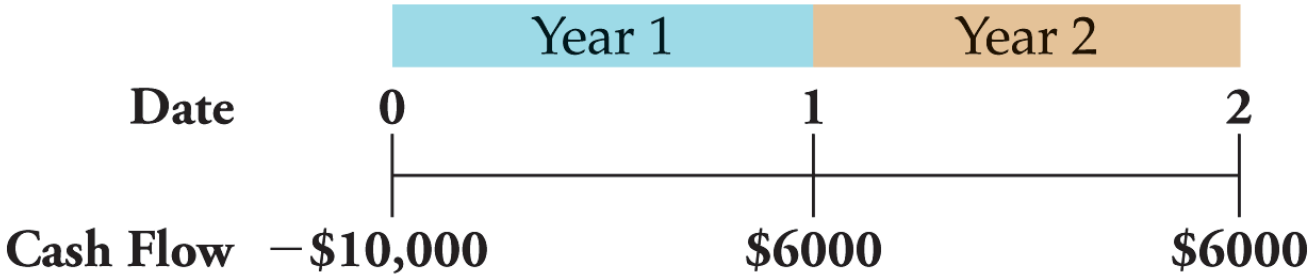
4.1 The Timeline (cont'd)

- Assume that you are lending $10,000 today and that the loan will be repaid in two annual $6,000 payments.



- The first cash flow at date 0 (today) is represented as a negative sum because it is an outflow.
- Timelines can represent cash flows that take place at the end of any time period – a month, a week, a day, etc.

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Textbook Example 4.1

Constructing a Timeline

Problem

Suppose you must pay tuition of $10,000 per year for the next two years. Your tuition payments must be made in equal installments at the start of each semester. What is the timeline of your tuition payments?

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Textbook Example 4.1 (cont'd)

Solution

Assuming today is the start of the first semester, your first payment occurs at date 0 (today). The remaining payments occur at semester intervals. Using one semester as the period length, we can construct a timeline as follows:

Date (Semesters)	0	1	2	3	4
Cash Flow	−$5000	−$5000	−$5000	−$5000	$0

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4.2 Three Rules of Time Travel

- Financial decisions often require combining cash flows or comparing values. Three rules govern these processes.

Table 4.1 The Three Rules of Time Travel

Rule 1	Only values at the same point in time can be compared or combined.	
Rule 2	To move a cash flow forward in time, you must compound it.	Future Value of a Cash Flow $FV_n = C \times (1 + r)^n$
Rule 3	To move a cash flow backward in time, you must discount it.	Present Value of a Cash Flow $PV = C \div (1+r)^n = \frac{C}{(1+r)^n}$

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The 1st Rule of Time Travel

- A dollar today and a dollar in one year are not equivalent.
- It is only possible to compare or combine values at the same point in time.
  - Which would you prefer: A gift of $1,000 today or $1,210 at a later date?
  - To answer this, you will have to compare the alternatives to decide which is worth more. One factor to consider: How long is "later?"

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The 2nd Rule of Time Travel

- To move a cash flow forward in time, you must compound it.
  - Suppose you have a choice between receiving $1,000 today or $1,210 in two years. You believe you can earn 10% on the $1,000 today, but want to know what the $1,000 will be worth in two years. The time line looks like this:

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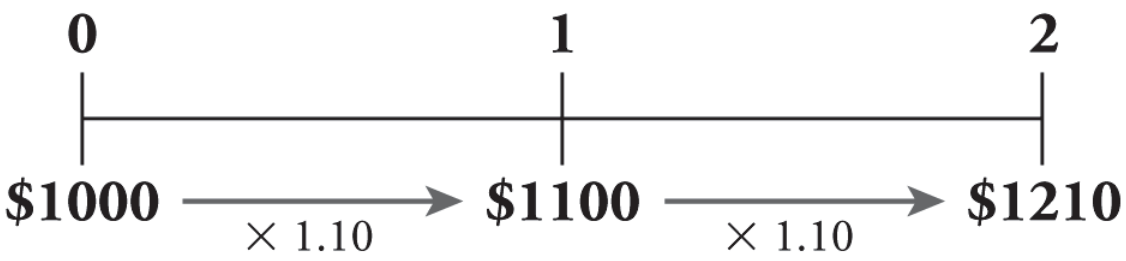
The 2nd Rule of Time Travel (cont'd)



- Future Value of a Cash Flow

$$FV_n = C \times \underbrace{(1 + r) \times (1 + r) \times \cdots \times (1 + r)}_{n \text{ times}} = C \times (1 + r)^n$$

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Using a Financial Calculator: The Basics

- BA-II Plus
  - Future Value FV
  - Present Value PV
  - I/Y I/Y
    - Interest Rate per Year
    - Interest is entered as a percent, not a decimal
      - For 10%, enter 10, NOT .10

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Using a Financial Calculator: The Basics (cont'd)

- BA-II Plus
  - Number of Periods N
  - Periodic Payment PMT
  - CLR TVM 2ND FV
    - Clears out all TVM registers
    - Should do between all problems

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Financial Calculator Solution

- Inputs:
  - $N = 2$
  - $I = 10$
  - $PV = 1{,}000$
- Output:
  - $FV = -1{,}210$

2	10	1,000		
N	I/Y	PV	PMT	FV
				-1,210

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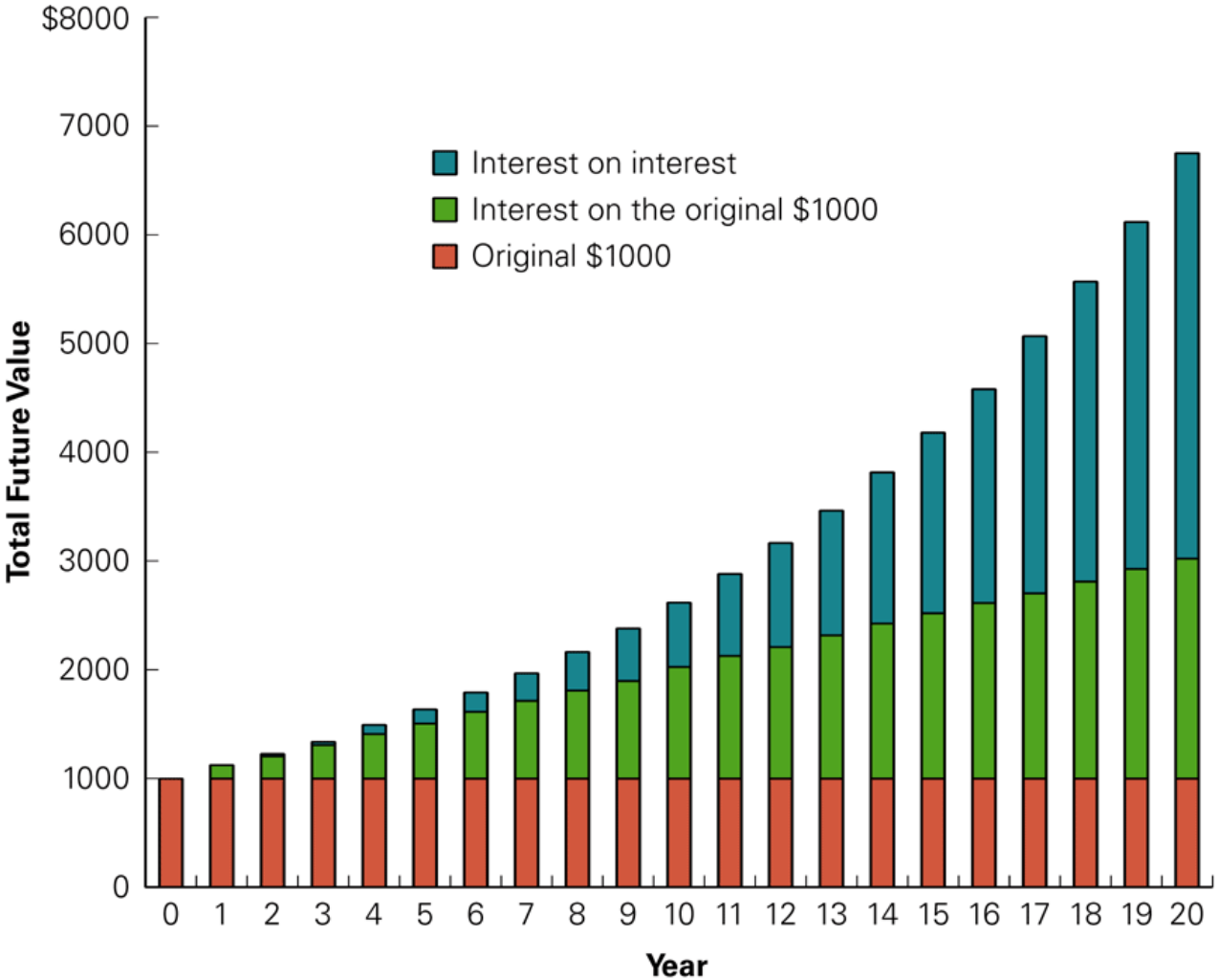
Figure 4.1 The Composition of Interest Over Time



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Textbook Example 4.2
Financial Calculator Solution for n=7 years

- Inputs:
  - $N = 7$
  - $I = 10$
  - $PV = 1{,}000$
- Output:
  - $FV = -1{,}948.72$

7	10	1,000		
N	I/Y	PV	PMT	FV
				-1,948.72

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The 3rd Rule of Time Travel

- To move a cash flow backward in time, we must discount it.

- Present Value of a Cash Flow

$$PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n}$$

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4.3 Valuing a Stream of Cash Flows

- Based on the first rule of time travel we can derive a general formula for valuing a stream of cash flows: if we want to find the present value of a stream of cash flows, we simply add up the present values of each.

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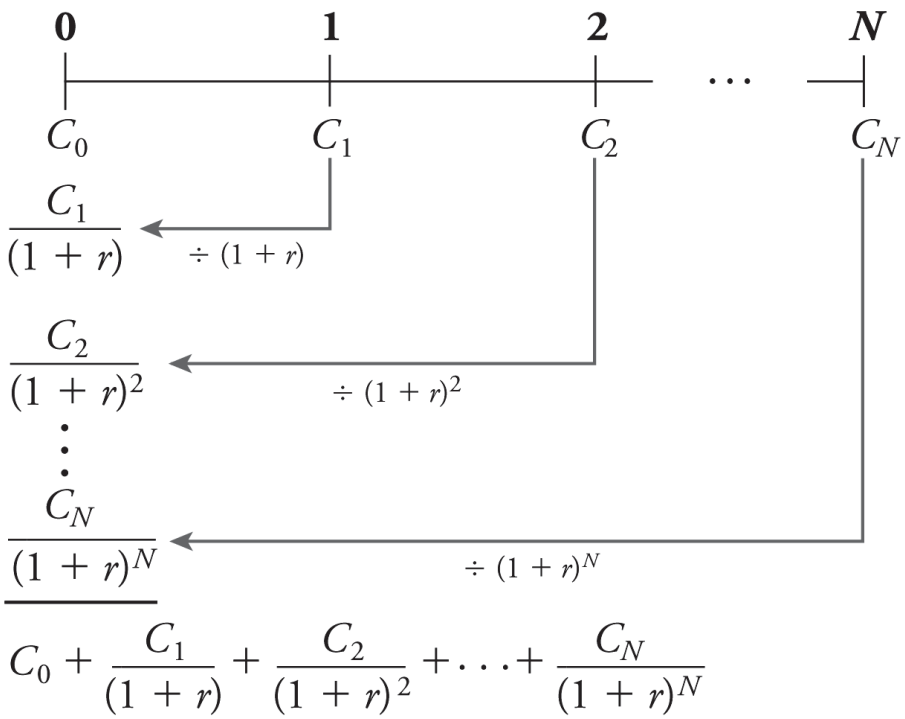
4.3 Valuing a Stream of Cash Flows (cont'd)



- Present Value of a Cash Flow Stream

$$PV = \sum_{n=0}^{N} PV(C_n) = \sum_{n=0}^{N} \frac{C_n}{(1+r)^n}$$

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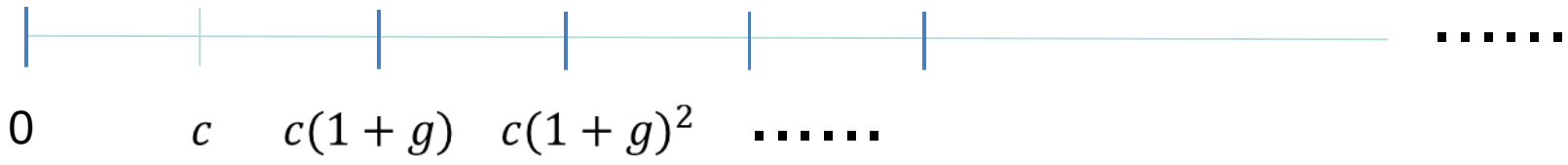
PV of Growing Annuity



$+$

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① $g = 0, n = \infty \quad PV = \frac{c}{r}$

② $g = 0, n \neq \infty \quad PV = \frac{c}{r}\left[1 - \frac{1}{(1+r)^n}\right]$

③ $g \neq 0, r > g, n \to \infty \quad PV = \frac{c}{r-g}$

④ $FV\ of\ growing\ annuity = ②(1+r)^n = \frac{c}{r}[(1+r)^n - 1]$

$Note: a + ar + ar^2 + \cdots + ar^n = a(1 + r + r^2 + \cdots + r^n)$

$$S^n = \frac{a(r^n - 1)}{r - 1}, n \to \infty, \mathrm{r} < 1 \Rightarrow S^n = \frac{a}{1 - r}$$

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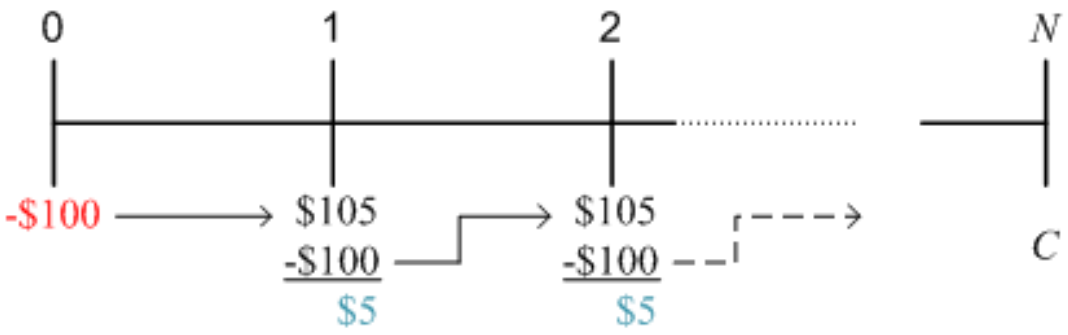
4.5 Perpetuities, Annuities, and Other Special Cases

- When a constant cash flow will occur at regular intervals forever it is called a perpetuity.



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4.5 Perpetuities, Annuities, and Other Special Cases (cont'd)

- The value of a perpetuity is simply the cash flow divided by the interest rate.
- Present Value of a Perpetuity

$$PV(C \text{ in perpetuity}) = \frac{C}{r}$$

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Textbook Example 4.7---No.1 Perpetuity

Endowing a Perpetuity

Problem

You want to endow an annual MBA graduation party at your alma mater. You want the event to be a memorable one, so you budget $30,000 per year forever for the party. If the university earns 8% per year on its investments, and if the first party is in one year's time, how much will you need to donate to endow the party?

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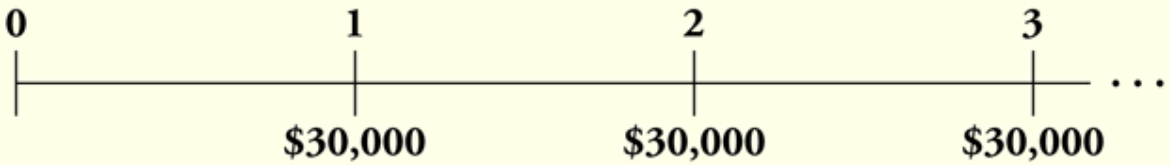
Textbook Example 4.7 (cont'd)

Solution

The timeline of the cash flows you want to provide is



This is a standard perpetuity of $30,000 per year. The funding you would need to give the university in perpetuity is the present value of this cash flow stream. From the formula,

$$PV = C/r = \$30{,}000/0.08 = \$375{,}000 \text{ today}$$

If you donate $375,000 today, and if the university invests it at 8% per year forever, then the MBAs will have $30,000 every year for their graduation party.

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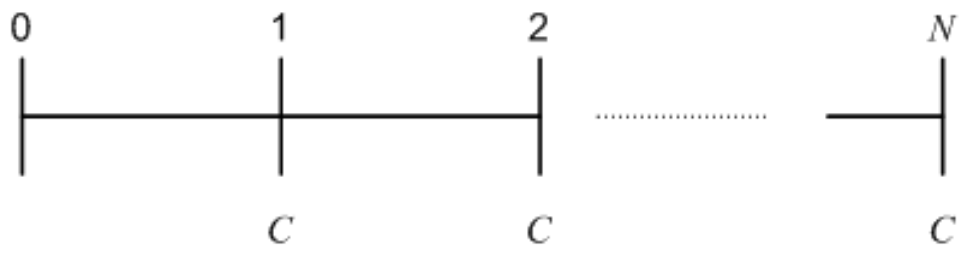
Annuities

- When a constant cash flow will occur at regular intervals for a finite number of *N* periods, it is called an annuity.



- Present Value of an Annuity

$$PV = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \frac{C}{(1+r)^3} + \ldots + \frac{C}{(1+r)^N} = \sum_{n=1}^{N} \frac{C}{(1+r)^n}$$

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Textbook Example 4.8-No.2

Present Value of a Lottery Prize Annuity

Problem

You are the lucky winner of the $30 million state lottery. You can take your prize money either as (a) 30 payments of $1 million per year (starting today), or (b) $15 million paid today. If the interest rate is 8%, which option should you take?

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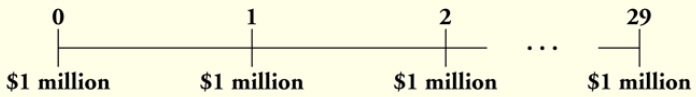
Textbook Example 4.8

Solution

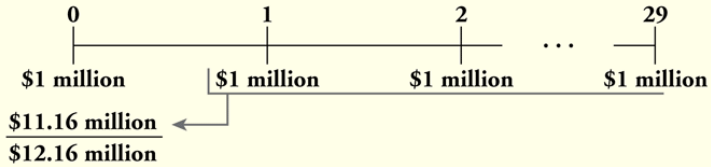
Option (a) provides $30 million in prize money but paid over time. To evaluate it correctly, we must convert it to a present value. Here is the timeline:



Because the first payment starts today, the last payment will occur in 29 years (for a total of 30 payments).[5] The $1 million at date 0 is already stated in present value terms, but we need to compute the present value of the remaining payments. Fortunately, this case looks like a 29-year annuity of $1 million per year, so we can use the annuity formula:

$$\begin{aligned} PV(\text{29-year annuity of \$1 million}) &= \$1\text{ million} \times \frac{1}{0.08}\left(1 - \frac{1}{1.08^{29}}\right) \\ &= \$1\text{ million} \times 11.16 \\ &= \$11.16\text{ million today} \end{aligned}$$

Thus, the total present value of the cash flows is $1 million + $11.16 million = $12.16 million. In timeline form:



Option (b), $15 million upfront, is more valuable—even though the total amount of money paid is half that of option (a). The reason for the difference is the time value of money. If you have the $15 million today, you can use $1 million immediately and invest the remaining $14 million at an 8% interest rate. This strategy will give you $14 million × 8% = $1.12 million per year in perpetuity! Alternatively, you can spend $15 million − $11.16 million = $3.84 million today, and invest the remaining $11.16 million, which will still allow you to withdraw $1 million each year for the next 29 years before your account is depleted.

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Textbook Example 4.10---No.3 Growing Perpetuity

Endowing a Growing Perpetuity

Problem

In Example 4.7, you planned to donate money to your alma mater to fund an annual $30,000 MBA graduation party. Given an interest rate of 8% per year, the required donation was the present value of

$$PV = \$30{,}000/0.08 = \$375{,}000 \text{ today}$$

Before accepting the money, however, the MBA student association has asked that you increase the donation to account for the effect of inflation on the cost of the party in future years. Although $30,000 is adequate for next year's party, the students estimate that the party's cost will rise by 4% per year thereafter. To satisfy their request, how much do you need to donate now?

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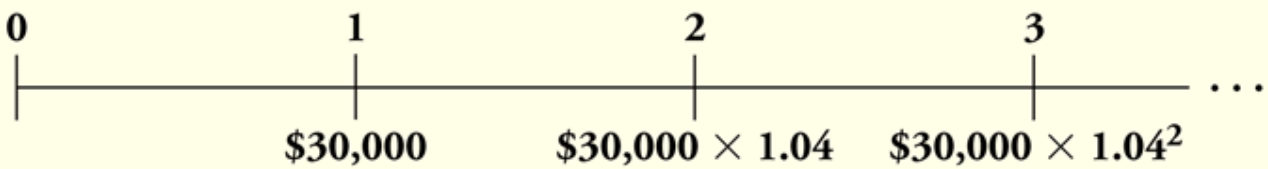
Textbook Example 4.10 (cont'd)

Solution



The cost of the party next year is $30,000, and the cost then increases 4% per year forever. From the timeline, we recognize the form of a growing perpetuity. To finance the growing cost, you need to provide the present value today of

$$PV = \$30{,}000/(0.08 - 0.04) = \$750{,}000 \text{ today}$$

You need to double the size of your gift!

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Textbook Example 4.9---No.4 FV of Annuity

Retirement Savings Plan Annuity

Problem

Ellen is 35 years old, and she has decided it is time to plan seriously for her retirement. At the end of each year until she is 65, she will save $10,000 in a retirement account. If the account earns 10% per year, how much will Ellen have saved at age 65?

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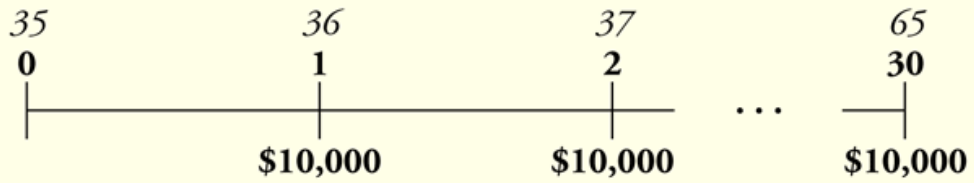
Textbook Example 4.9 (cont'd)

Solution

As always, we begin with a timeline. In this case, it is helpful to keep track of both the dates and Ellen's age:



Ellen's savings plan looks like an annuity of $10,000 per year for 30 years. (*Hint*: It is easy to become confused when you just look at age, rather than at both dates and age. A common error is to think there are only 65 − 36 = 29 payments. Writing down both dates and age avoids this problem.)

To determine the amount Ellen will have in the bank at age 65, we compute the future value of this annuity:

$$
\begin{aligned}
FV &= \$10{,}000 \times \frac{1}{0.10}(1.10^{30} - 1) \\
&= \$10{,}000 \times 164.49 \\
&= \$1.645 \text{ million at age 65}
\end{aligned}
$$

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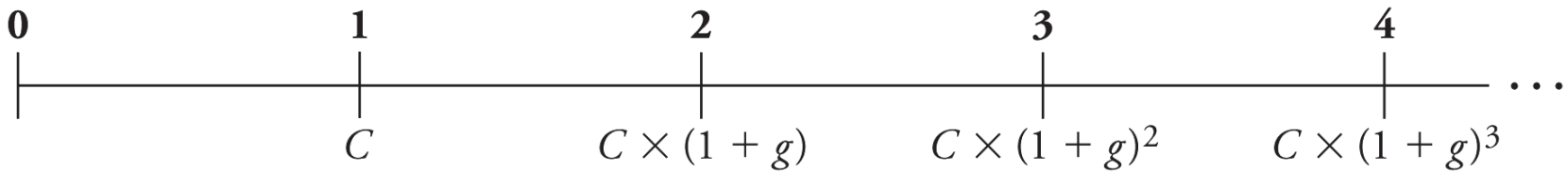
Growing Perpetuities

- Assume you expect the amount of your perpetual payment to increase at a constant rate, *g*.



- Present Value of a Growing Perpetuity

$$PV\ (\text{growing perpetuity}) = \frac{C}{r - g}$$

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Growing Annuities

- The present value of a growing annuity with the initial cash flow *c*, growth rate *g*, and interest rate *r* is defined as:
  - Present Value of a Growing Annuity

$$PV = C \times \frac{1}{(r - g)} \left(1 - \left(\frac{1 + g}{(1 + r)} \right)^N \right)$$

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Textbook Example 4.11---Mix FV of Growing Annuities

Retirement Savings with a Growing Annuity

Problem

In Example 4.9, Ellen considered saving $10,000 per year for her retirement. Although $10,000 is the most she can save in the first year, she expects her salary to increase each year so that she will be able to increase her savings by 5% per year. With this plan, if she earns 10% per year on her savings, how much will Ellen have saved at age 65?

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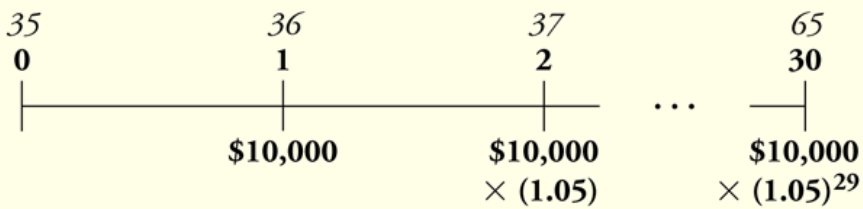
Textbook Example 4.11

Solution

Her new savings plan is represented by the following timeline:



This example involves a 30-year growing annuity, with a growth rate of 5%, and an initial cash flow of $10,000. The present value of this growing annuity is given by

$$PV = \$10{,}000 \times \frac{1}{0.10 - 0.05}\left(1 - \left(\frac{1.05}{1.10}\right)^{30}\right)$$

$$= \$10{,}000 \times 15.0463$$

$$= \$150{,}463 \text{ today}$$

Ellen's proposed savings plan is equivalent to having $150,463 in the bank *today*. To determine the amount she will have at age 65, we need to move this amount forward 30 years:

$$FV = \$150{,}463 \times 1.10^{30}$$

$$= \$2.625 \text{ million in 30 years}$$

Ellen will have saved $2.625 million at age 65 using the new savings plan. This sum is almost $1 million more than she had without the additional annual increases in savings.

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Textbook Example 4.13

Using the Annuity Spreadsheet

Problem

Suppose that you invest $20,000 in an account paying 8% interest. You plan to withdraw $2000 at the end of each year for 15 years. How much money will be left in the account after 15 years?

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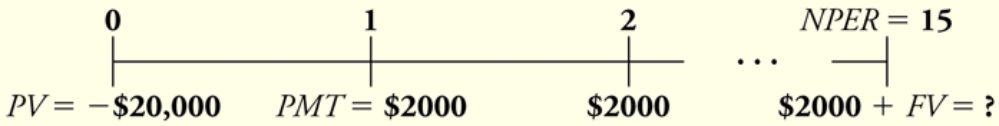
Textbook Example 4.13 (cont'd)

Solution

Again, we start with the timeline:



The timeline indicates that the withdrawals are an annuity payment that we receive from the bank account. Note that *PV* is negative (money *into* the bank), while *PMT* is positive (money *out* of the bank). We solve for the final balance in the account, *FV*, using the annuity spreadsheet:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	15	8.00%	−20,000	2000		
Solve for FV					**9139**	=FV(0.08,15,2000,−20000)

We will have $9139 left in the bank after 15 years.

We can also compute this solution directly. One approach is to think of the deposit and the withdrawals as being separate accounts. In the account with the $20,000 deposit, our savings will grow to $63,443 in 15 years, as we computed in Example 4.12. Using the formula for the future value of an annuity, if we borrow $2000 per year for 15 years at 8%, at the end our debt will have grown to

$$\$2000 \times \frac{1}{0.08}(1.08^{15} - 1) = \$54{,}304$$

After paying off our debt, we will have $63,443 − $54,304 = $9139 remaining after 15 years.

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4.7 Solving for Variables Other Than Present Values or Future Values (cont'd)

- In some situations, you know the present value and cash flows of an investment opportunity but you do not know the **internal rate of return (IRR)**, the interest rate that sets the net present value of the cash flows equal to zero.

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Textbook Example 4.15

Computing the Internal Rate of Return with the Annuity Spreadsheet in Excel

Problem

Jessica has just graduated with her MBA. Rather than take the job she was offered at a prestigious investment bank—Baker, Bellingham, and Botts—she has decided to go into business for herself. However, Baker, Bellingham, and Botts was so impressed with Jessica that it has decided to fund her business. In return for an initial investment of $1 million, Jessica has agreed to pay the bank $125,000 at the end of each year for the next 30 years. What is the internal rate of return on Baker, Bellingham, and Botts's investment in Jessica's company, assuming she fulfills her commitment?

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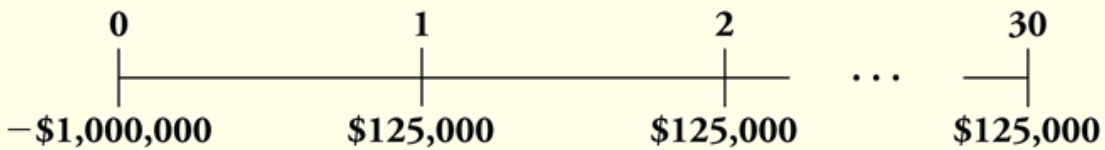
Textbook Example 4.15 (cont'd)

Solution

Here is the timeline (from Baker, Bellingham, and Botts's perspective):



The timeline shows that the future cash flows are a 30-year annuity. Setting the NPV equal to zero requires

$$1{,}000{,}000 = 125{,}000 \times \frac{1}{r}\left(1 - \frac{1}{(1+r)^{30}}\right)$$

Using the annuity spreadsheet to solve for r,

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	30		−1,000,000	125,000	0	
Solve for Rate		**12.09%**				=RATE(30,125000,−1000000,0)

The IRR on this investment is 12.09%.

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Chapter Quiz

1. Can you compare or combine cash flows at different times?
2. How do you calculate the present value of a cash flow stream?
3. What benefit does a firm receive when it accepts a project with a positive NPV?
4. How do you calculate the present value of a
   a. Perpetuity?
   b. Annuity?
   c. Growing perpetuity?
   d. Growing annuity?

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